

4(a)(i)	frequency or period	B1
4(a)(ii)	amplitude	B1
4(b)	constant phase difference so coherent	B1
4(c)	120°	B1
4(d)	resultant displacement = $4.0\ \mu\text{m} - 1.0\ \mu\text{m}$ = $3.0\ \mu\text{m}$	B1
4(e)	$I \propto A^2$	C1
	intensity of Z = $(2^2 / 4^2) I$ = $0.25 I$	A1
4(f)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1 / T$	C1
	$330 = \lambda / 3.0 \times 10^{-3}$	C1
	$\lambda = 0.99\ \text{m}$	A1